Name: ____________________________ (   ) **CT Group: 22S0** _______

RAFFLES INSTITUTION

2022 YEAR 6 TERM 2 TIMED PRACTICE

22 March 2022 Time: 2 hr

H2 PHYSICS

Part 1

INSTRUCTIONS TO CANDIDATES

There are 2 parts in this paper.

Part 1 consists of **5** structured questions.

Part 2 consists of **2** longer structured questions.

Attempt ALL questions in Parts 1 and 2.

Write your name, index number and CT Group on the cover page.

Write your answers to Part 1 in the spaces provided on the question paper.

For Examiner's Use		
Part 1	1	/ 9
	2	/ 7
	3	/ 11
	4	/ 9
	5	/ 12
Part 2	6	/ 16
	7	/ 16
Deductions		
Total		/ 80
%		

This document consists of **12** printed pages.

Data

speed of light in free space	c	=	3.00×10^{8} m s^{-1}
permeability of free space	μ_0	=	$4\pi \times 10^{-7}$ H m^{-1}
permittivity of free space	ε_0	=	8.85×10^{-12} F m^{-1}
		=	$(1/(36\pi)) \times 10^{-9}$ F m^{-1}
elementary charge	e	=	1.60×10^{-19} C
the Planck constant	h	=	6.63×10^{-34} J s
unified atomic mass constant	u	=	1.66×10^{-27} kg
rest mass of electron	m_e	=	9.11×10^{-31} kg
rest mass of proton	m_p	=	1.67×10^{-27} kg
molar gas constant	R	=	8.31 J K^{-1} mol^{-1}
the Avogadro constant	N_A	=	6.02×10^{23} mol^{-1}
the Boltzmann constant	k	=	1.38×10^{-23} J K^{-1}
gravitational constant	G	=	6.67×10^{-11} N m^{2} kg^{-2}
acceleration of free fall	g	=	9.81 m s^{-2}

Formulae

uniformly accelerated motion	s	=	$ut + \frac{1}{2}at^2$
	v^2	=	$u^2 + 2as$
work done on/by a gas	W	=	$p\Delta V$
hydrostatic pressure	p	=	ρgh
gravitational potential	ϕ	=	$-Gm/r$
temperature	T/K	=	T / °C + 273.15
pressure of an ideal gas	p	=	$\frac{1}{3}\frac{Nm}{V}\langle c^2 \rangle$
mean translational kinetic energy of an ideal gas molecule	E	=	$\frac{3}{2}kT$
displacement of particle in s.h.m.	x	=	$x_0 \sin \omega t$
velocity of particle in s.h.m.	v	=	$v_0 \cos \omega t = \pm \omega \sqrt{x_0^2 - x^2}$
electric current	I	=	$Anvq$
resistors in series	R	=	$R_1 + R_2 + \ldots$
resistors in parallel	$1/R$	=	$1/R_1 + 1/R_2 + \ldots$
electric potential	V	=	$\frac{Q}{4\pi\varepsilon_0 r}$
alternating current/voltage	x	=	$x_0 \sin \omega t$
magnetic flux density due to a long straight wire	B	=	$\frac{\mu_0 I}{2\pi d}$
magnetic flux density due to a flat circular coil	B	=	$\frac{\mu_0 NI}{2r}$
magnetic flux density due to a long solenoid	B	=	$\mu_0 nI$
radioactive decay	x	=	$x_0 \exp(-\lambda t)$
decay constant	λ	=	$\ln 2 / t_{1/2}$

Part 1 (48 marks)

1 **(a)** Using Newton's law of gravitation and the definition of gravitational field strength, derive an expression for the gravitational field strength g at a distance R from a point mass of mass M. Explain your working.

[2]

(b) Neutron stars are the remnants of massive stars that collapse after a supernova explosion in space. They are said to be the densest form of observable matter in the universe.

The Crab Nebular Pulsar is a neutron star with a mass 1.5 times and a radius 10^{-5} times that of our Sun.

(i) Given that the gravitational field strength at the surface of our Sun is 290 N kg^{-1}, calculate the gravitational field strength at the surface of the Crab Nebular Pulsar.

gravitational field strength = ... N kg^{-1} [2]

(ii) The Crab Nebular Pulsar has a radius of 1.1×10^4 m and rotates about its axis with a period of 0.24 s.

1. Calculate the centripetal acceleration of a particle moving in a circular path of radius 1.1×10^4 m with a period of rotation of 0.24 s.

centripetal acceleration = ... m s^{-2} [2]

2. Hence deduce whether particles on the surface of the Crab Nebular Pulsar will leave its surface due to the high speed of rotation of the star.

....

....

....

.... [2]

(iii) Explain why Neutron stars, like the Crab Nebular Pulsar, have extremely smooth surfaces.

....

.... [1]

2 A point source of light, operating at constant power, shines light uniformly in all directions. A detector is placed in front of the light source along a central axis and at a distance 5.0 m away as shown in Fig. 2.1.

The intensity of the light incident on the detector is measured to be I_0.

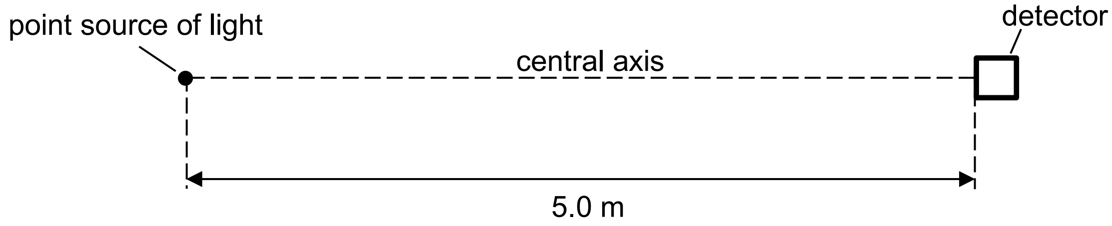

Fig. 2.1

(a) When the detector is moved away from the source by a distance *x* along the central axis, the intensity measured by the detector decreases to $\frac{I_0}{3}$.

Determine *x*.

x = ... m [2]

(b) The detector is moved back to its initial position and a student places a thin sheet of polarising filter directly in front of the detector as shown in Fig. 2.2.

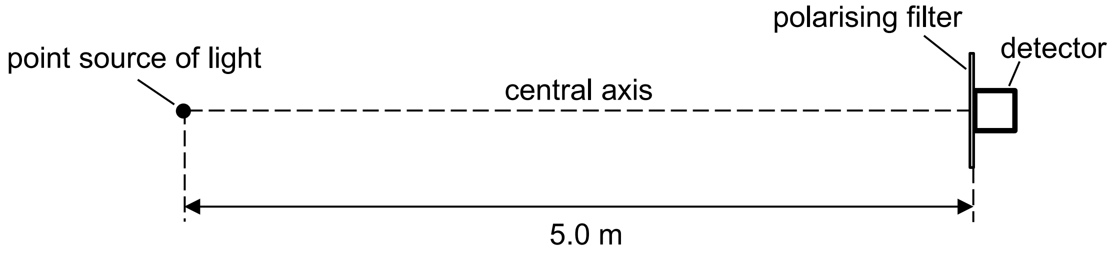

Fig. 2.2

The intensity measured by the detector then decreases to $\frac{I_0}{5}$.

(i) Explain why this intensity value implies that the light from the light source is already polarised before passing through the polarising filter.

....

.... [1]

(ii) Determine the angle between the plane of polarisation of the light from the source and the polarising axis of the polarising filter.

angle = ... ° [2]

(c) The polarising filter is removed and a reflector is placed directly behind the light source as shown in Fig. 2.3. The detector, at 5.0 m away from the light source, measures the intensity to be I_1.

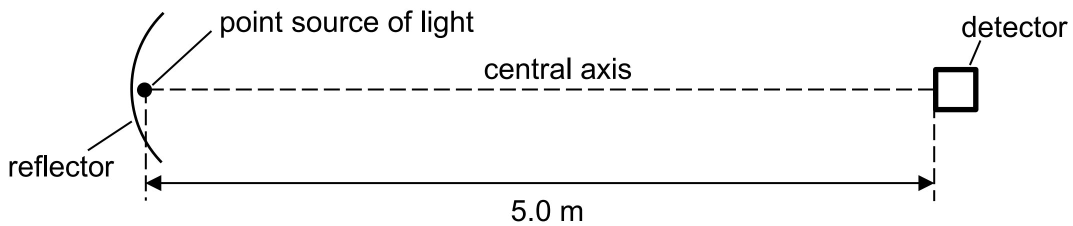

Fig. 2.3

With reference to the definition of the intensity of a wave, state and explain whether I_1 is larger than, smaller than, or equal to I_0.

....

....

.... [2]

3 **(a)** State the *first law of thermodynamics* in words.

....

....

.... [2]

(b) A fixed mass of monatomic ideal gas undergoes a cycle of changes $A \rightarrow B \rightarrow C \rightarrow A$ as shown in Fig. 3.1.

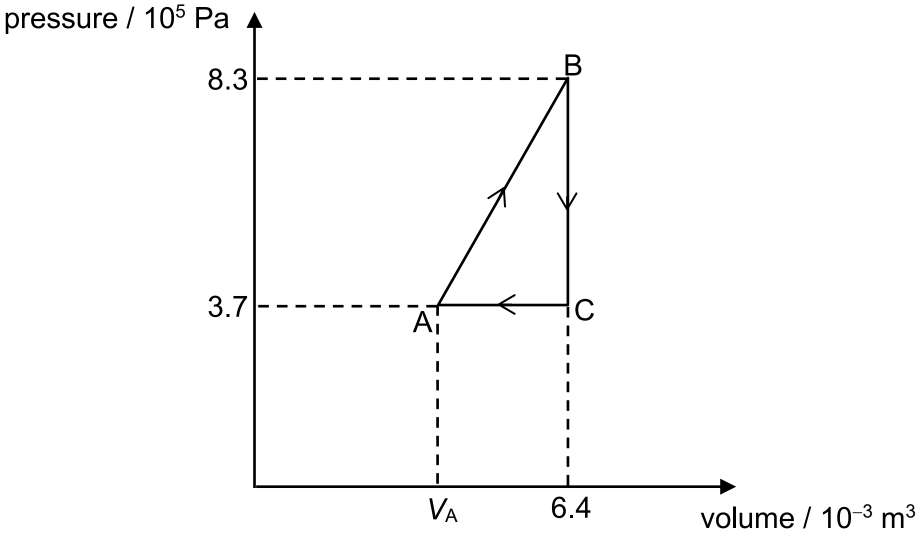

Fig. 3.1

The temperature of the gas at A and B are 130 K and 500 K respectively.

(i) Show that the volume V_A of the gas at A is 3.7×10^{-3} m^3.

[2]

(ii) Deduce, with clear explanations, whether heat is supplied to or taken out of the gas during the change $A \rightarrow B$.

....

....

....

.... [2]

(iii) Calculate the change in internal energy during the change $B \rightarrow C$.

change in internal energy = ... J [2]

(iv) Calculate the net heat supplied to the gas in the cycle of changes $A \rightarrow B \rightarrow C \rightarrow A$.

net heat supplied = ... J [3]

4 Two parallel metal plates, each of length 7.5 cm, are separated by a distance of 3.6 cm. The potential difference of 900 V between the plates produces a uniform electric field in the region between the plates.

A particle of charge $-2e$ and mass 6.6×10^{-27} kg has an initial speed of 4.1×10^{5} m s^{-1}. It is travelling parallel to the metal plates and enters the electric field mid-way between the plates.

The whole setup, as shown in Fig. 4.1, is in vacuum.

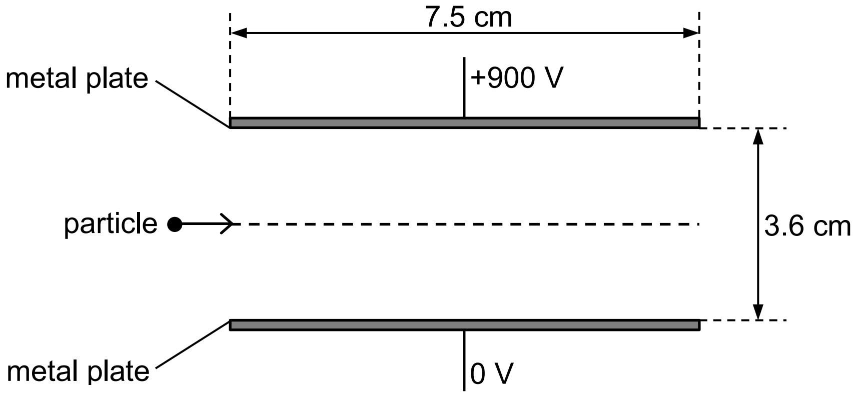

Fig. 4.1

(a) State what is meant by a *uniform electric field*.

....

.... [1]

(b) (i) On Fig. 4.1, draw the electric field lines between the plates. [1]

(ii) State the direction of the electric force on the charged particle when it is between the plates.

.... [1]

(c) Consider the charged particle when it is travelling between the plates.

(i) Show that the acceleration of the particle due to the electric field is 1.21×10^{12} m s^{-2}.

[1]

(ii) Hence show, with appropriate calculations and explanations, that the particle collides with one of the plates.

....

....

.... [3]

(iii) Sketch on Fig. 4.2, how the particle's electric potential energy U varies with its vertical displacement d from its point of entry until it collides with the plate.

Include appropriate values on the axes.

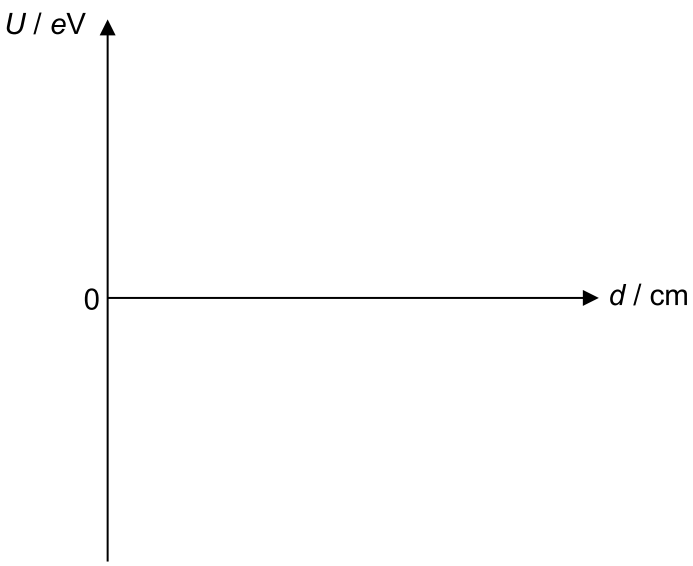

Fig. 4.2

[2]

5 A dry-cell of e.m.f. E and internal resistance r of 0.162 Ω is connected in series to a switch S, a variable resistor of resistance R and an ideal ammeter as shown in Fig. 5.1. An ideal voltmeter is connected across the dry-cell.

When switch S is opened, the voltmeter reads 1.50 V.

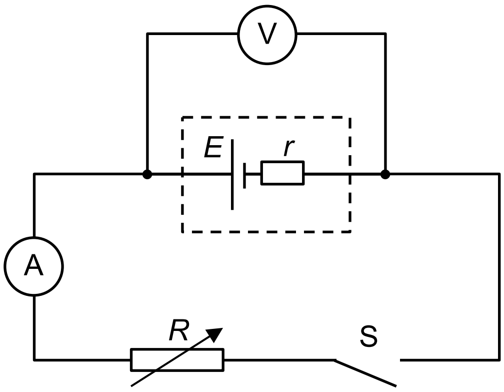

Fig. 5.1

(a) When the variable resistor is set to its maximum resistance and switch S is closed, the voltmeter reads 1.20 V.

(i) Show that the maximum resistance of the rheostat is 0.648 Ω.

[2]

(ii) Calculate the current measured by the ammeter.

current = ... A [1]

(iii) On Fig. 5.2, draw a graph to show the variation with the current I measured by the ammeter of the voltmeter reading V as the resistance R of the variable resistor is reduced from its maximum value to zero.

Include the following in your graph:

1. Label, with their coordinates, the points on your graph that correspond to the maximum value of R and when R is zero.

2. Indicate the value of the e.m.f. E in relation to your graph.

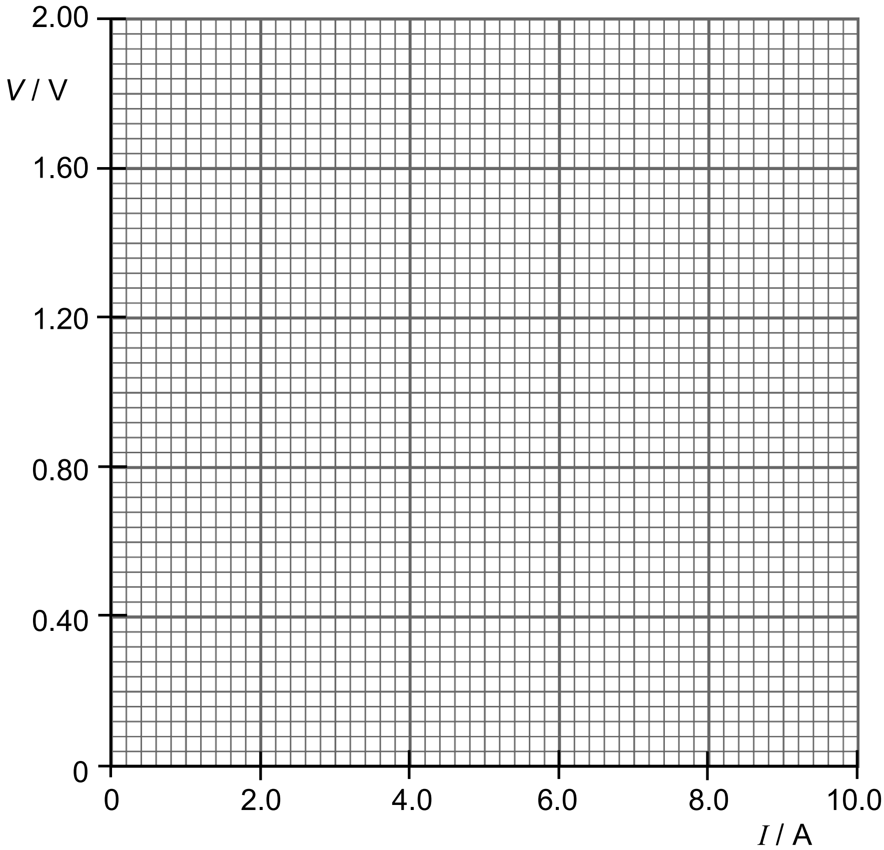

Fig. 5.2

[2]

(iv) The graph in **(a)(iii)** is now drawn for a similar circuit but with a non-ideal ammeter.

State, with reasons, two similarities and one difference between this graph for a non-ideal ammeter and the graph that you have drawn in **(a)(iii)** for an ideal ammeter.

Similarities:

1. .... [1]

2. .... [1]

Difference:

1. ....

.... [1]

(b) The power P dissipated in the variable resistor varies with its resistance R.

(i) Determine the maximum power dissipated in the variable resistor.

maximum power = ... W [2]

(ii) On Fig. 5.3, draw a graph to show the variation with R of P as the resistance of the variable resistor is varied from 0 Ω to 0.648 Ω.

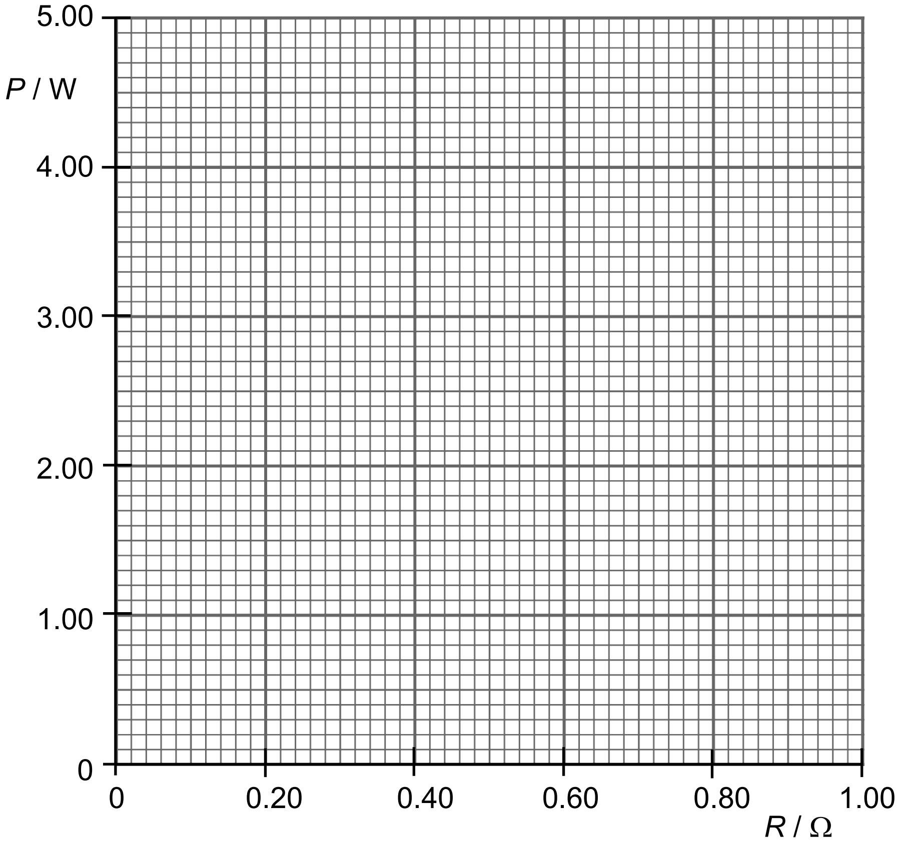

Fig. 5.3

[2]

END OF PART 1